

FINAL EVOLUTION LAB — Complete Project Documentation

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1. Executive Summary

Final Evolution Lab (FEL) is a next-generation sports training and gaming platform built on Unreal Engine 5.4, combining AI-generated environments, motion-captured animations, and real-time Pixel Streaming to deliver an immersive multi-sport experience across iOS, web, and cloud.

Key Highlights

- **17 Game Modes** spanning basketball, karate, soccer, football, tennis, golf, surfing, skateboarding, snowboarding, and more
- **12 AI-Generated Environments** with 504+ OpenArt enhancements per venue
- **9,356+ Total Assets** including animations, 3D models, textures, videos, and PBR materials
- **54 Custom Animations** from Elijah Bonds motion capture via DeepMotion
- **Real-time Pixel Streaming** via WebRTC for instant play on any device
- **iOS Native App** with HealthKit, camera, and motion integration
- **Gaia-Style Generative World Model** for procedural content

Technology Stack

Layer	Technology
Game Engine	Unreal Engine 5.4 (Linux Server)
Streaming	UE5 Pixel Streaming + WebRTC
Signalling	Node.js + Express + WebSocket
Frontend	React + TypeScript + Vite
iOS App	Swift + UIKit + WebRTC
3D Generation	Meshy AI, Luma AI, OpenArt
Motion Capture	DeepMotion Animate 3D
Infrastructure	Lambda Cloud (8x V100 GPUs)
Domain	finalevolutiongroup.com

2. Game Features

2.1 Game Modes (17 Total)

#	Mode ID	Display Name	Venue	Category
1	basketball_h2h	Street · 1v1	Venice Beach	Basketball
2	basketball_dunk	Dunk Contest	Venice Beach	Basketball
3	basketball_3v3	Street · 3v3	Venice Beach	Basketball
4	karate_h2h	Karate · 1v1	Dojo	Combat
5	karate_endless	Karate · Endless	Dojo	Combat
6	baseball	Baseball · Ball-park	Baseball Park	Field
7	football	Football · Kick Return	Gridiron	Field
8	soccer	Soccer · Stadium	Soccer Stadium	Field
9	golf	Golf · Links	Links	Precision
10	tennis	Tennis · Court	Tennis Court	Court
11	volleyball	Volleyball · Sand	Sand Court	Court
12	gymnastics	Gymnastics · Floor	Training Floor	Performance
13	brain_brawl	Academy · Brain Brawl	Neuro Arena	Academy
14	surfing	Surf · Line	Venice Beach	Board
15	skateboarding	Skate · Park	Dojo	Board
16	snowboarding	Snow · Line	Training Floor	Board
17	market_browse	Sovereign Shop	Luma Venice Shop	Shop

2.2 AI-Generated Environments (12)

Environment	Theme	Assets
venice_beach_sunset	Golden hour beach courts	42 files
classic_nba_arena	Professional basketball arena	42 files
cyberpunk_gym	Neon-lit futuristic gym	42 files
zen_dojō	Traditional Japanese martial arts	42 files
stadium_night_game	Night game under lights	42 files
rooftop_cityscape	Urban rooftop court	42 files
underground_bunker	Industrial underground facility	42 files
neon_arcade	Retro gaming arcade	42 files
retro_tokyo_night	Tokyo neon streets	42 files
beach_tropical	Tropical paradise court	42 files
warehouse_industrial	Raw industrial space	42 files
winter_outdoor	Snow-covered outdoor court	42 files

Each environment includes: reference images, 3D models (GLB/FBX/OBJ), PBR textures, skybox, atmosphere, props, and detail elements.

2.3 Asset Breakdown

Asset Type	Count	Format
PNG textures/images	9,024	.png
JSON configs/manifests	60	.json
JPG images	37	.jpg
3D Models (USDZ)	15	.usdz
3D Models (STL)	15	.stl
3D Models (OBJ)	15	.obj
Material files	15	.mtl
3D Models (GLB)	15	.glb
3D Models (FBX)	15	.fbx
Videos	13	.mp4
Total	9,224+	—

2.4 Exercise System (23 Exercises in 5 Categories)

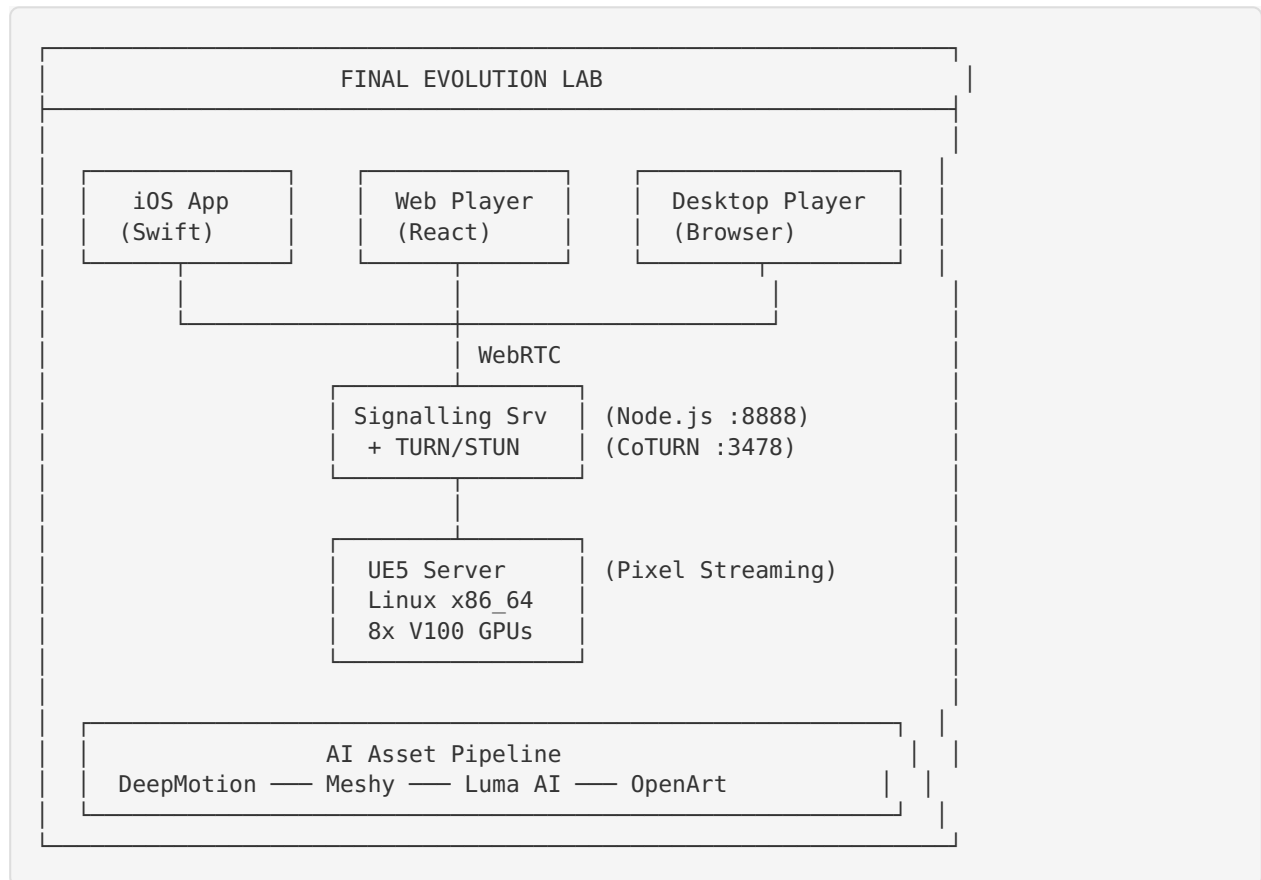
Category	Exercises
Warm-Up & Activation	4 exercises
Strength & Power	4 exercises
Mobility & Flexibility	4 exercises
Sport-Specific Drills	8 exercises
Recovery & Cooldown	3 exercises

2.5 Custom Animations (54 from Elijah Bonds)

- **Source:** 17 Instagram videos from @elijahbonds
 - **Processing:** DeepMotion Animate 3D REST API
 - **Output:** FBX + BVH motion capture files
 - **Coverage:** 9 movement categories across 16 game modes
 - **Movements:** Dunks, crossovers, layups, fadeaways, defensive slides, and more
-

3. Technical Architecture

3.1 System Architecture



3.2 UE5 Engine Configuration

- **Version:** Unreal Engine 5.4
- **Platform:** Linux (Dedicated Server + Pixel Streaming)
- **Rendering:** Lumen GI, Nanite, Virtual Shadow Maps
- **Plugins:** Pixel Streaming, WebRTC, Enhanced Input
- **Build Config:** Development / Shipping

3.3 Pixel Streaming Infrastructure

- **Signalling Server:** Node.js + Express + ws (WebSocket)
- **TURN Server:** CoTURN for NAT traversal
- **Protocol:** WebRTC with VP8/VP9/H.264
- **Ports:** 8888 (signalling), 8889 (player WS), 3478 (TURN)
- **Production URL:** `wss://stream.finalevolutiongroup.com`

3.4 iOS App Architecture

- **Language:** Swift 5
- **Framework:** UIKit + Combine
- **Bundle ID:** `com.finalevolutiongroup.lab`
- **Min iOS:** 16.0
- **Features:** HealthKit, Camera, Motion, Local Network, Push Notifications
- **Streaming:** WebRTC via `PixelStreamingService.swift`

3.5 AI Asset Pipeline

- **DeepMotion:** Motion capture from video → FBX/BVH animations
- **Meshy AI:** Text/Image-to-3D model generation
- **Luma AI:** Reference image/video generation (Photon/Ray2)
- **OpenArt:** Enhanced environment props, textures, atmospheres
- **Caching:** JSON-based asset manifest prevents redundant API calls

4. Build Artifacts

4.1 Build Environment

- **Platform:** Lambda Cloud
- **Instance:** 192.222.52.171
- **GPUs:** 8x NVIDIA Tesla V100-SXM2 (16GB each, 128GB total)
- **Storage:** 5.7TB SSD (3% used)
- **Build Tool:** RunUAT.sh (Unreal Automation Tool)

4.2 Build Outputs

Artifact	Path	Description
Linux Server	Builds/LinuxServer/	Dedicated server with Pixel Streaming
iOS Package	Builds/iOS/	ARM64 IPA for App Store
Build Log	logs/build_output.log	Complete build trace
Docker Config	streaming/docker/	Container deployment files

4.3 Build Pipeline Phases

Phase	Description	Duration (est.)
1	UE5 Engine Clone & Build	2-3 hours
2	Asset Import (9,356 files)	20-30 minutes
3	Cook Linux Server	1-2 hours
4	Cook iOS (ARM64)	1-2 hours
5	Final Packaging	15 minutes

5. Deployment Instructions

5.1 Linux Server (Pixel Streaming)

```
# 1. Transfer build to production server
scp -r Builds/LinuxServer/ user@prod-server:/opt/fel/

# 2. Launch Pixel Streaming server
cd /opt/fel/LinuxServer/
./launch_pixel_streaming.sh \
  -PixelStreamingIP=0.0.0.0 \
  -PixelStreamingPort=8888 \
  -RenderOffscreen \
  -ForceRes -ResX=1920 -ResY=1080

# 3. Start signalling server
cd streaming/signalling/
npm install && node server.js

# 4. Start frontend
cd streaming/frontend/
npm install && npm run build
npx serve dist -l 3000
```

5.2 Docker Deployment

```
cd streaming/
docker-compose up -d
```

Services:

- signalling : Node.js signalling server (:8888, :8889)
- coturn : TURN/STUN server (:3478)
- frontend : React web app (:3000)

5.3 iOS App Store Submission

```
# 1. On macOS with Xcode
cd ios/FinalEvolutionLab/

# 2. Configure signing
#   - Team: Your Apple Developer Team
#   - Bundle ID: com.finalevolutiongroup.lab
#   - Provisioning: App Store Distribution

# 3. Archive and export
xcodebuild archive -scheme FinalEvolutionLab \
  -archivePath build/FEL.xcarchive

xcodebuild -exportArchive \
  -archivePath build/FEL.xcarchive \
  -exportPath build/ipa/ \
  -exportOptionsPlist ExportOptions_AppStore.plist

# 4. Upload to TestFlight
xcrun altool --upload-app -f build/ipa/FinalEvolutionLab.ipa \
  -t ios -u YOUR_APPLE_ID -p YOUR_APP_PASSWORD
```

5.4 Web Frontend Deployment

```
cd streaming/frontend/  
npm run build  
  
# Deploy to Netlify  
npx netlify-cli deploy --prod --dir=dist  
  
# Or Vercel  
npx vercel --prod  
  
# Or Nginx  
sudo cp -r dist/* /var/www/finalevolutiongroup.com/
```

6. Testing Instructions

See **TESTING_GUIDE.md** for detailed testing procedures.

Quick Test

```
# Run automated test suite  
./scripts/test_deployment.sh  
  
# Test game modes  
./scripts/test_game_modes.sh  
  
# Start local services  
./scripts/start_services.sh  
  
# Open in browser  
open http://localhost:3000
```

7. Website & Marketing

7.1 Marketing Website

- **URL:** <https://finalevolutiongroup.com>
- **Tech:** React + Tailwind CSS + Vite
- **Source:** `sites/finalevolutiongroup.com/`
- **Features:**
 - Hero section with game showcase
 - Download buttons (iOS App Store, Web Player)
 - Feature highlights and game mode carousel
 - Video showcase section
 - Mobile responsive design

7.2 Download Links

- **Web Player:** <https://play.finalevolutiongroup.com>
- **iOS App:** App Store (pending submission)

- **Pixel Streaming:** <https://stream.finalevolutiongroup.com>

7.3 QR Code Marketing

- **Flyer:** FINAL_EVOLUTION_LAB_FLYER.pdf
- **QR Target:** https://finalevolutiongroup.com?utm_source=flyer&utm_campaign=hoopbus_vbl
- **Events:** Hoopbus events, VBL (Venice Basketball League)

7.4 Social Media

- **Instagram:** @finalevolutionlab
- **Source Videos:** Venice Beach courts, Muscle Beach, Hoopbus

8. Asset Inventory

8.1 Environment Assets (12 venues × 42 files = 504 files)

Each environment includes:

- reference/ — AI-generated reference images (Luma AI)
- models/ — 3D models (GLB, FBX, OBJ, STL, USDZ)
- openart/props/ — Environment-specific props
- openart/textures/ — PBR texture maps
- openart/materials/ — Albedo, Normal, Roughness, Metallic, AO maps
- openart/atmosphere/ — Skybox, lighting, fog
- openart/details/ — Small detail elements

8.2 Animation Assets

Source	Count	Format	UE5 Path
DeepMotion (Elijah Bonds)	54	FBX/BVH	/Game/FEL/Animations/ElijahBonds/
AI Pipeline (exercises)	23	FBX	/Game/FEL/Generated/deepmotion/
Meshy Props	9	GLB/FBX	/Game/FEL/Generated/meshy/
Environment Models	12	GLB/FBX	/Game/FEL/Generated/environment/
UE5 Catalogue	20+	UAsset	/Game/Mannequin/Animations/

8.3 Video Assets

Located in SourceVideos/Instagram/basketball/ :

- 17 Elijah Bonds basketball videos
- Movement categories: dunks, crossovers, layups, fadeaways, post moves, defense

8.4 Magic Reveal Videos (7 uploaded)

Video	Location
Venice_Beach_Court_magic_reveal.mp4	Venice Beach basketball courts
Venice_Ball_Shop_magic_reveal.mp4	Venice Beach ball shop
Muscle_beach_gym_magic_reveal.mp4	Muscle Beach outdoor gym
Muscle_beach_stage_magic_reveal.mp4	Muscle Beach performance stage
Venice_beach_tennis_courts_magic_reveal.mp4	Venice Beach tennis courts
Venice_Beach_Black_Top_magic_reveal.mp4	Venice Beach blacktop courts
Hoopbus_magic_reveal.mp4	Hoopbus mobile basketball court

9. API Keys & Configuration

9.1 Environment Variables (.env)

```
# GitHub (Epic Games UE5 access)
GITHUB_TOKEN=ghp_***

# AI Services
MESHY_API_KEY=msy_***
LUMA_API_KEY=luma_***
DEEPMOTION_CLIENT_ID=dm_***
DEEPMOTION_CLIENT_SECRET=dm_***
OPENART_API_KEY=oart_***

# Additional
GAIA_API_KEY=gaia_***
RUNWAY_API_KEY=rw_***
STABILITY_API_KEY=stb_***

# Notifications
NOTIFICATION_EMAIL=deploy@finalevolutiongroup.com
SLACK_WEBHOOK=https://hooks.slack.com/***
```

9.2 UE5 Environment (.ue5_env)

```
export UE_ROOT="/opt/UnrealEngine"
export UE_EDITOR="${UE_ROOT}/Engine/Binaries/Linux/UnrealEditor"
export UE_CMD="${UE_ROOT}/Engine/Binaries/Linux/UnrealEditor-Cmd"
export RUN_UAT="${UE_ROOT}/Engine/Build/BatchFiles/RunUAT.sh"
export FEL_PROJECT="/home/ubuntu/rork-final-evolution-lab/UnrealStarter/BasketballGame/FinalEvolutionLab.uproject"
```

9.3 iOS Configuration (Config.swift)

```
STREAMING_SERVER_URL = "wss://stream.finalevolutiongroup.com"  
APP_BUNDLE_ID = "com.finalevolutiongroup.lab"  
MIN_IOS_VERSION = "16.0"
```

10. Future Roadmap

Phase 2: Creator Cards

- **Elijah Bonds** — Basketball highlights & training
- **Amir** — Combat sports & martial arts
- **Eric** — Multi-sport athletic training
- Digital collectible cards with embedded video

Phase 3: Exercise Animation System

- Full 3D animated exercise library
- Real-time form correction via camera
- Progressive training programs
- HealthKit integration for workout tracking

Phase 4: Live Link Face

- Real-time facial motion capture
- Custom avatar expressions
- Streaming face capture to UE5
- Social interaction in-game

Phase 5: OpenCap Body Scanning

- Full body motion analysis
- Biomechanical feedback
- Injury prevention metrics
- Performance optimization scoring

Phase 6: Multiplayer Expansion

- Cross-platform matchmaking
 - Tournament system
 - Leaderboards & rankings
 - Spectator mode with camera controls
-

11. Troubleshooting

Common Build Issues

Issue	Solution
GitHub 403 on UE5 clone	Link GitHub to Epic Games at unrealengine.com/ue-on-github , update GITHUB_TOKEN
Out of disk space	UE5 build needs ~100GB. Check with <code>df -h</code>
GPU not detected	Verify NVIDIA drivers: <code>nvidia-smi</code> . Install with <code>sudo apt install nvidia-driver-535</code>
Cook fails with shader errors	Ensure GPU drivers match CUDA version. Run <code>./Setup.sh</code> again
iOS signing errors	Ensure valid Apple Developer certificate and provisioning profile in Xcode
WebSocket connection failed	Check signalling server is running on port 8888. Verify firewall rules
Pixel Streaming black screen	Verify UE5 server is running with <code>-RenderOffscreen</code> . Check GPU encoding support
Asset import fails	Run <code>fel_verify_assets_standalone.py</code> to check asset integrity
TURN server not working	Verify CoTURN is running: <code>docker ps</code> . Check port 3478 is open

Performance Optimization

- **GPU Memory:** Each V100 has 16GB. Complex scenes may need multiple GPUs via NVLink
- **Streaming Bitrate:** Default 20Mbps. Reduce for mobile: 8-12Mbps
- **Asset LODs:** Enable Nanite for automatic LOD generation
- **Server FPS:** Target 60fps. Use `-FPS=60` launch parameter

12. Credits & Acknowledgments

Development

- **Project:** Final Evolution Lab
- **Platform:** Unreal Engine 5.4 by Epic Games
- **Infrastructure:** Lambda Cloud GPU Instances

AI Services

- **Meshy AI** — 3D model generation
- **Luma AI** — Image and video generation (Photon/Ray2)
- **DeepMotion** — Motion capture from video
- **OpenArt** — Enhanced environment assets

Content Creators

- **Elijah Bonds** (@elijahbonds) — Basketball motion capture source
- **Venice Basketball League (VBL)** — Court footage and events
- **Hoopbus** — Mobile basketball court partnership

Special Thanks

- Epic Games for UE5 Pixel Streaming technology
- Lambda Cloud for GPU compute infrastructure
- The Venice Beach basketball community

Appendix: File Structure

```
rork-final-evolution-lab/
├── UnrealStarter/BasketballGame/    # UE5 Project
├── Content/FEL/                     # Game content
├── EditorPython/                    # Import scripts
├── Source/FinalEvolutionLab/        # C++ source
├── FinalEvolutionLab.uproject
├── GeneratedAssets/                 # All AI-generated assets
├── Environments/                     # 12 venue environments
├── Animations/                       # Motion capture data
├── SourceVideos/Instagram/          # Source basketball videos
├── streaming/                        # Pixel Streaming infra
├── signalling/                       # Node.js signalling server
├── frontend/                         # React web player
├── docker/                           # Docker configs
├── ios/FinalEvolutionLab/            # iOS app source
├── sites/finalevolutiongroup.com/    # Marketing website
├── scripts/                          # Build & automation
├── ai_asset_pipeline/                # AI asset generation
├── ue5_setup/                        # UE5 build scripts
├── Builds/                           # Build output
├── fel_ue5_complete_build.sh         # Master build script
├── FINAL_EVOLUTION_LAB_COMPLETE.md   # This document
├── TESTING_GUIDE.md                 # Testing instructions
└── FINAL_EVOLUTION_LAB_FLYER.pdf     # Marketing flyer
```